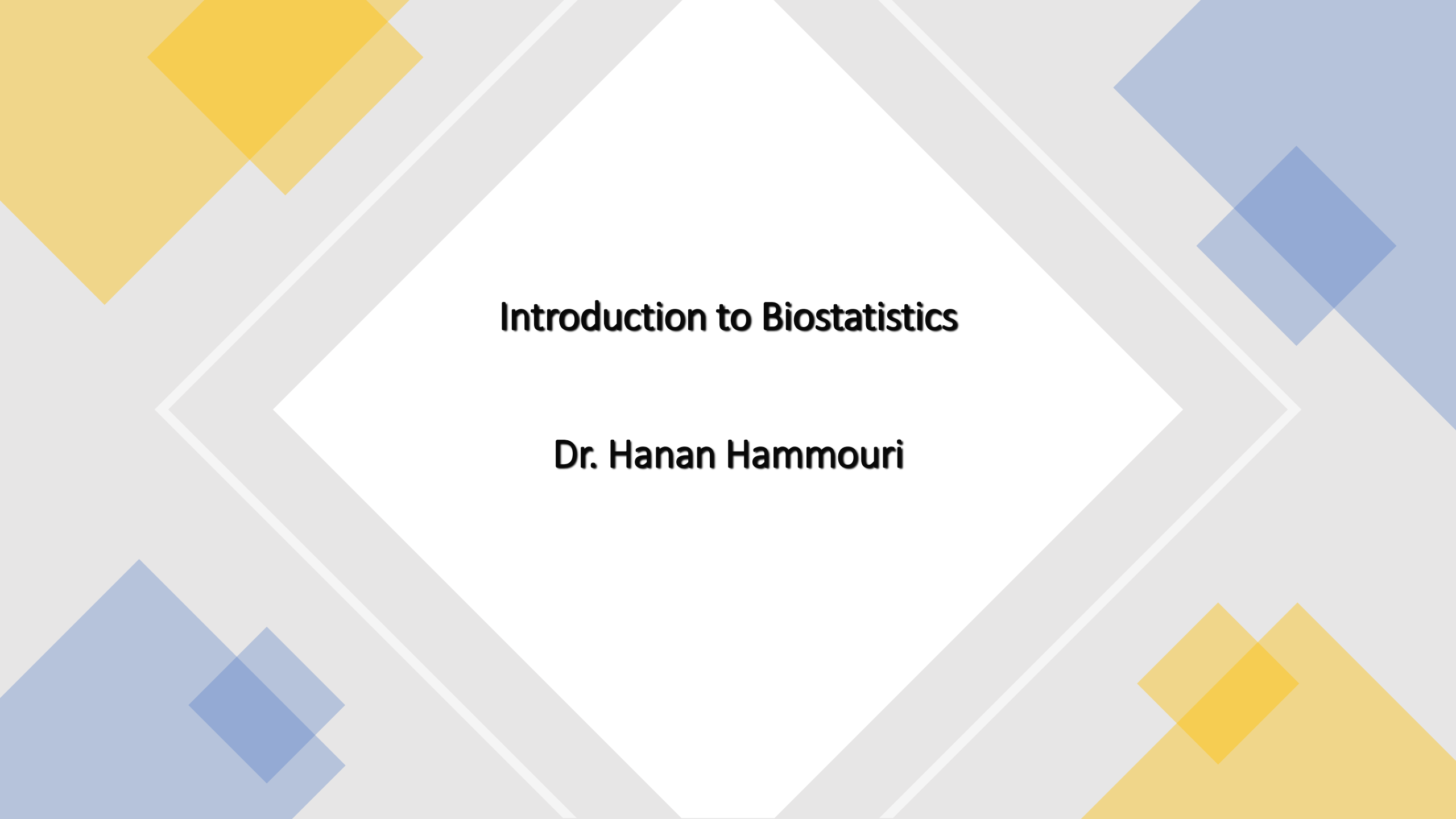




Introduction to Biostatistics

Dr. Hanan Hammouri

Biostatistics

- (a portmanteau word Biology and statistics merged into one word)
- The application of statistics to a wide range of topics in biology.

Biostatistics

The science concerned with developing and applying the most appropriate methods for:

- Data collection.
- Data presentation.
- Results analysis and interpretation.
- Analysis-based decisions.

RECOMMENDED TEXT BOOK

- Wayne W. Daniel, Biostatistics: A foundation for analysis in health sciences, 10th edition,

SOME BASIC CONCEPTS

- **Data:** The raw material of statistics, numbers we collect.
- **Statistics:** The field of study concerned with:
 - the collection, organization, summarization, and analysis of data.
 - the drawing of inferences about a body of data when only a part of the data is observed.

- **Sources of Data:**
 - Routinely kept records.
 - Surveys.
 - Experiments.
 - External sources.

- **Population:** a population of values of a variable,
Can be defined as the largest collection of values
of a random variable for which we have an
interest at a particular time.
- **Sample:** a sample may be defined simply as a
part of a population.

Types of statistical summaries

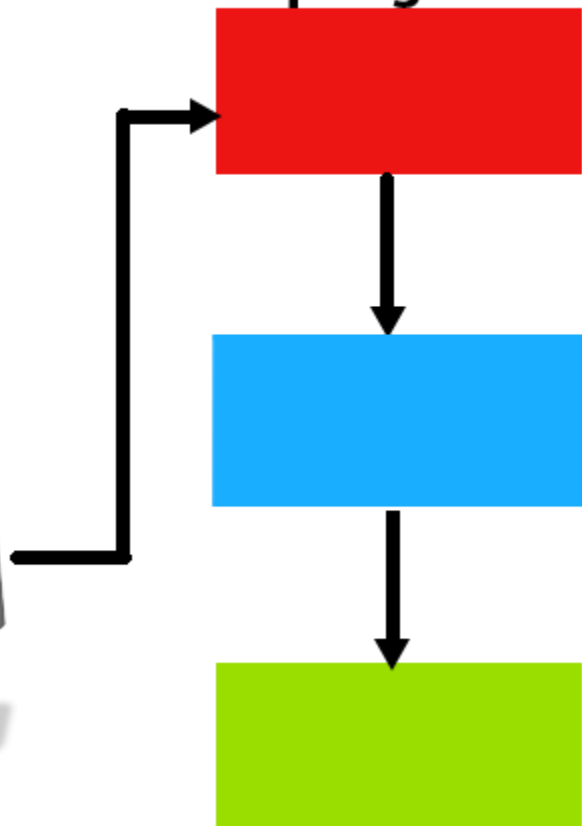
- **Parameter:** a data summary from a population
- **Statistic:** a data summary from a sample

<u>Parameter</u>	<u>Statistic</u>
μ : population mean	$\bar{x}$: sample mean
σ : population standard deviation	s : sample standard deviation
p : population proportion	$\hat{p}$: sample proportion

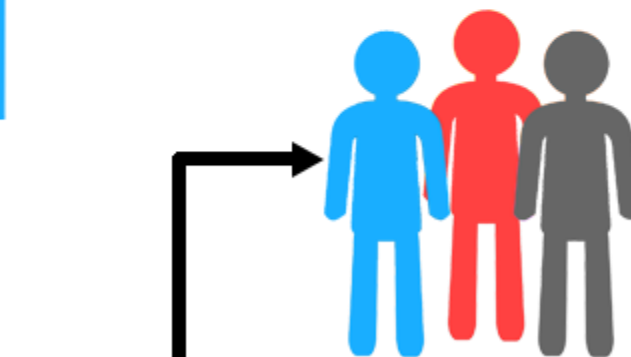
Population



Sampling Process



Sampling



Types of statistics

- **Descriptive statistics** – Methods of organizing, summarizing, and presenting data in an informative way
- **Inferential statistics** – The methods used to determine something about a population on the basis of a sample
 - Population –The entire set of individuals or objects of interest or the measurements obtained from all individuals or objects of interest
 - Sample – A portion, or part, of the population of interest

Data

- data are individual pieces of factual information recorded and used for the purpose of analysis. It is the raw information from which statistics are created.

	A	B	C	D	E	F	G
1	ID	Age	Gender	District	SATV	SATM	GPA
2	54419	18	M	38	368	253	3.52
3	62516	22	M	5	670	496	1.11
4	55509	21	F	54	639	439	2.68
5	36489	19	M	49	368	465	3.11
6	36387	21	F	36	620	306	2.16
7	95507	20	F	13	512	593	2.83
8	16360	20	M	52	621	377	2.79
9	12838	18	F	44	571	544	2.13
10	73450	20	F	59	647	746	2.08
11	26869	18	F	28	337	371	2.28
12	48552	22	M	63	260	498	3.24
13	23416	19	M	51	476	294	2.31
14	42635	19	F	35	677	241	3.19
15	67448	19	F	55	335	533	1.81
16	34689	21	F	42	585	708	1.80
17	32763	22	F	20	556	787	1.18

- **Variable:** it is an observed characteristic, that it takes on different values in different persons, places, or things.
 - Random Variable
 - discrete
 - continuous.

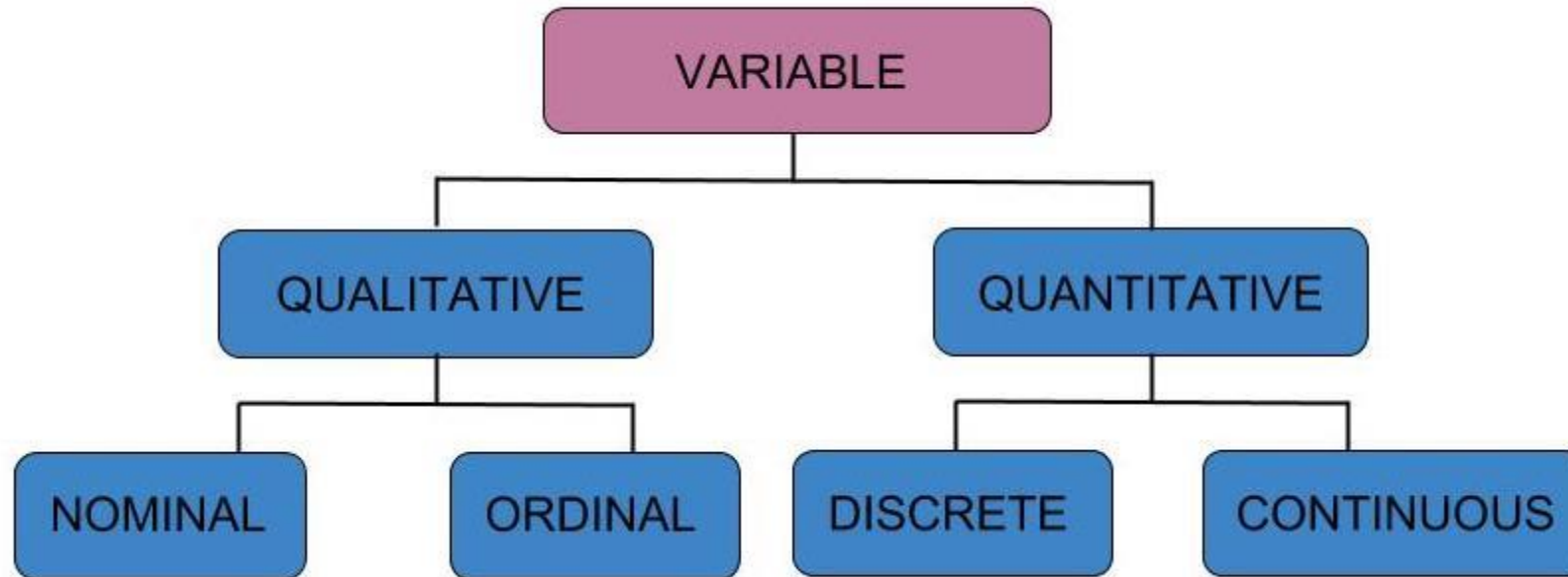
MEASUREMENT AND MEASUREMENT SCALES

- **Measurement:** the assignment of numbers to objects or events according to a set of rules.
 - **The Nominal Scale** (categorical): when the order is unimportant. (Male and Female for gender)
 - **The Ordinal Scale:** the order is important but not the difference between values. (Pain scale from 1 to 10.)
 - **The Interval Scale:** is a measurement where the difference between two values is meaningful. (temperature, height, weight)
 - **The Ratio Scale:** has all the properties of an interval variable and a clear definition of 0. When the variable equals 0, there is none of that variable. (height, weight, enzyme activity)

Operations that make sense for variables of different scales:

Scale	Operations that make sense			
	Counting	Ranking	Addition/ Subtraction	Multiplication/ Division
Nominal	✓			
Ordinal	✓	✓		
Interval	✓	✓	✓	
Ratio	✓	✓	✓	✓

Types of variables

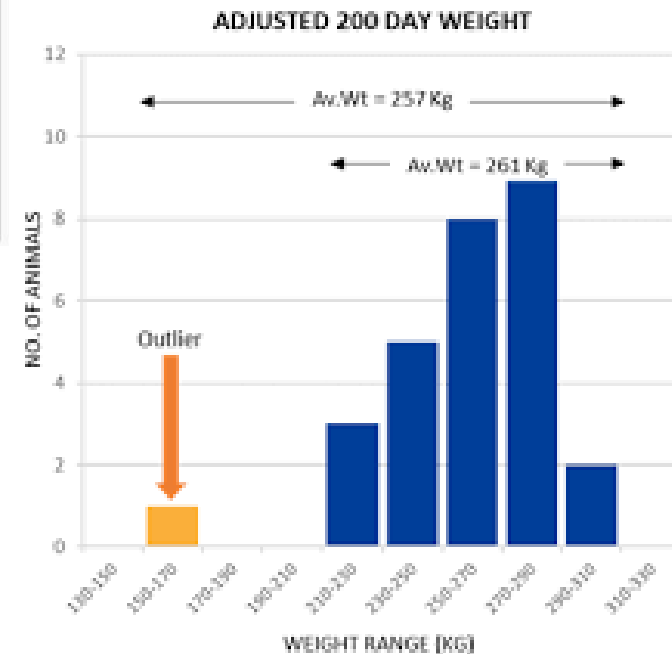
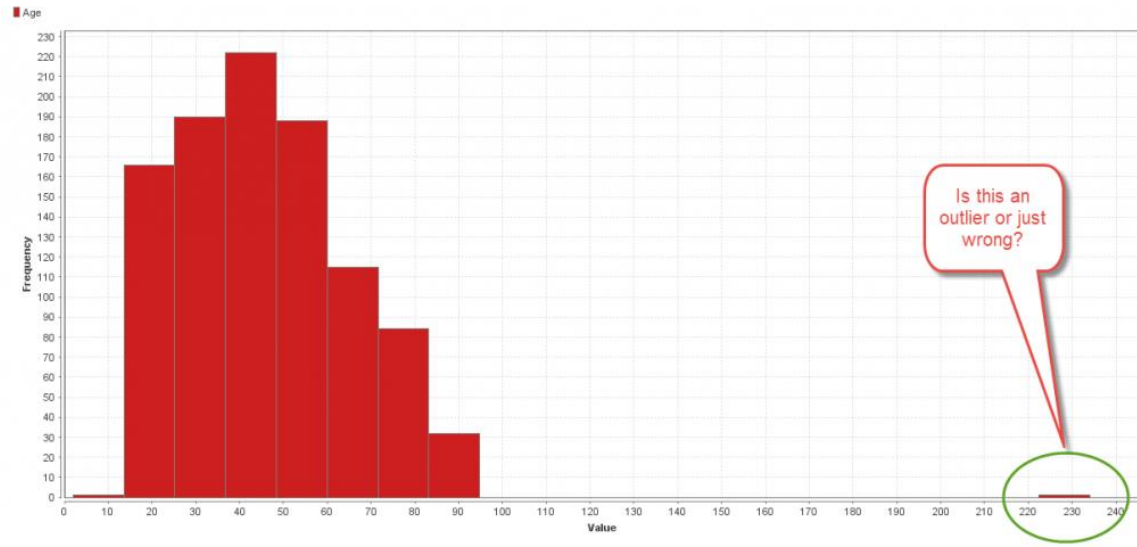


First part: Categorical or Qualitative Data

- **Graphs and Tables for Data Description:**
 - Tables and graphs, particularly of frequency distributions, are an important aspect of descriptive statistics.
 - Frequency Distribution Table
 - Histograms
 - Bar Charts
 - Pie Charts

Outlier

- An outlier is an observation that lies an abnormal distance from other values in a random sample



Category and frequency information is presented in a table (Frequency Distribution).

- Two types of frequency distributions that are most often used are the categorical frequency distribution and the grouped frequency distribution.

Table of Student Work Hours with Relative Frequencies		
DATA VALUE	FREQUENCY	RELATIVE FREQUENCY
2	3	0.15
3	5	0.25
4	3	0.15
5	6	0.3
6	2	0.1
7	1	0.05

Table represents the heights, in inches, of a sample of 100 male soccer players.

HEIGHTS (INCHES)	FREQUENCY	RELATIVE FREQUENCY	CUMULATIVE RELATIVE FREQUENCY
59.95–61.95	5	0.05	0.05
61.95–63.95	3	0.03	0.08
63.95–65.95	15	0.15	0.23
65.95–67.95	40	0.40	0.63
67.95–69.95	17	0.17	0.80
69.95–71.95	12	0.12	0.92
71.95–73.95	7	0.07	0.99
73.95–75.95	1	0.01	1.00
	Total = 100	Total = 1.00	

Charts and graphs

- Frequency distributions are good ways to present the essential aspects of data collections in concise and understandable terms
- Pictures are always more effective in displaying large data collections

Bar Graph

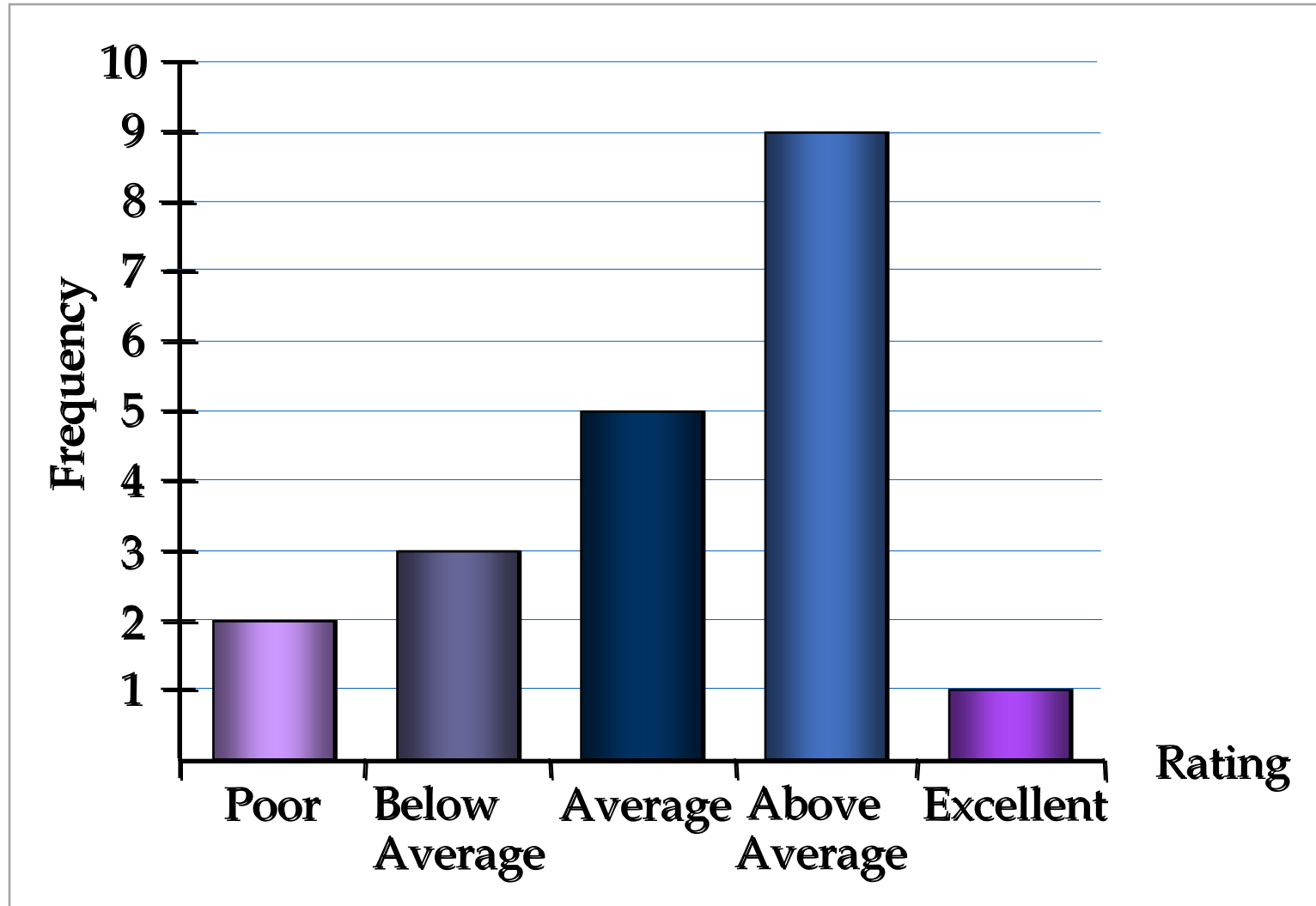
A bar graph is a graphical device for depicting qualitative data.

On one axis (usually the horizontal axis), we specify the labels that are used for each of the classes.

A frequency, relative frequency, or percent frequency scale can be used for the other axis (usually the vertical axis).

The bars are separated to emphasize the fact that each class is a separate category.

Bar Graph

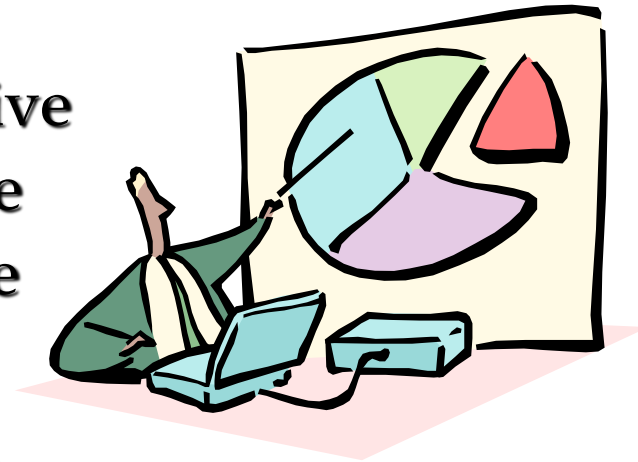


Pie Chart

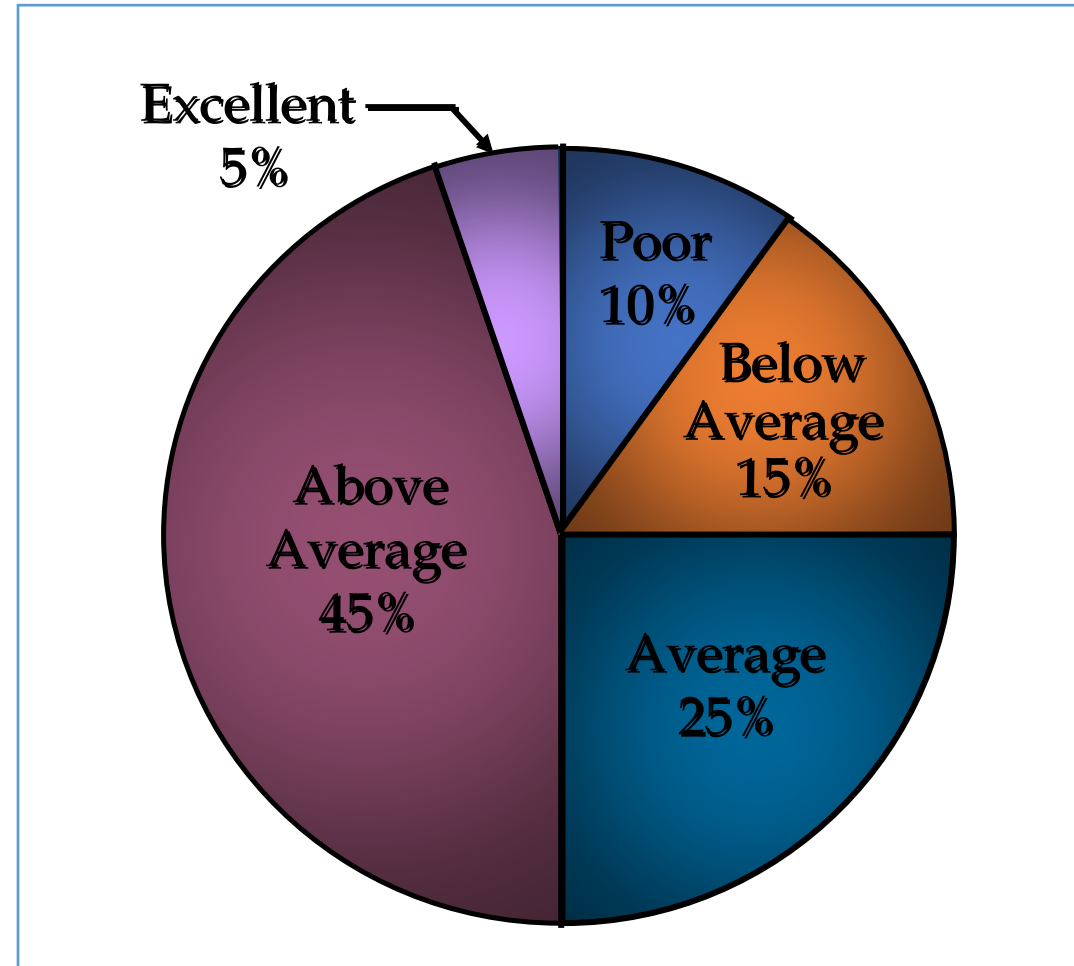
The pie chart is a commonly used graphical device for presenting relative frequency distributions for qualitative data.

First draw a circle; then use the relative frequencies to subdivide the circle into sectors that correspond to the relative frequency for each class.

Since there are 360 degrees in a circle, a class with a relative frequency of .25 would consume $.25(360) = 90$ degrees of the circle.



Pie Chart



Histogram

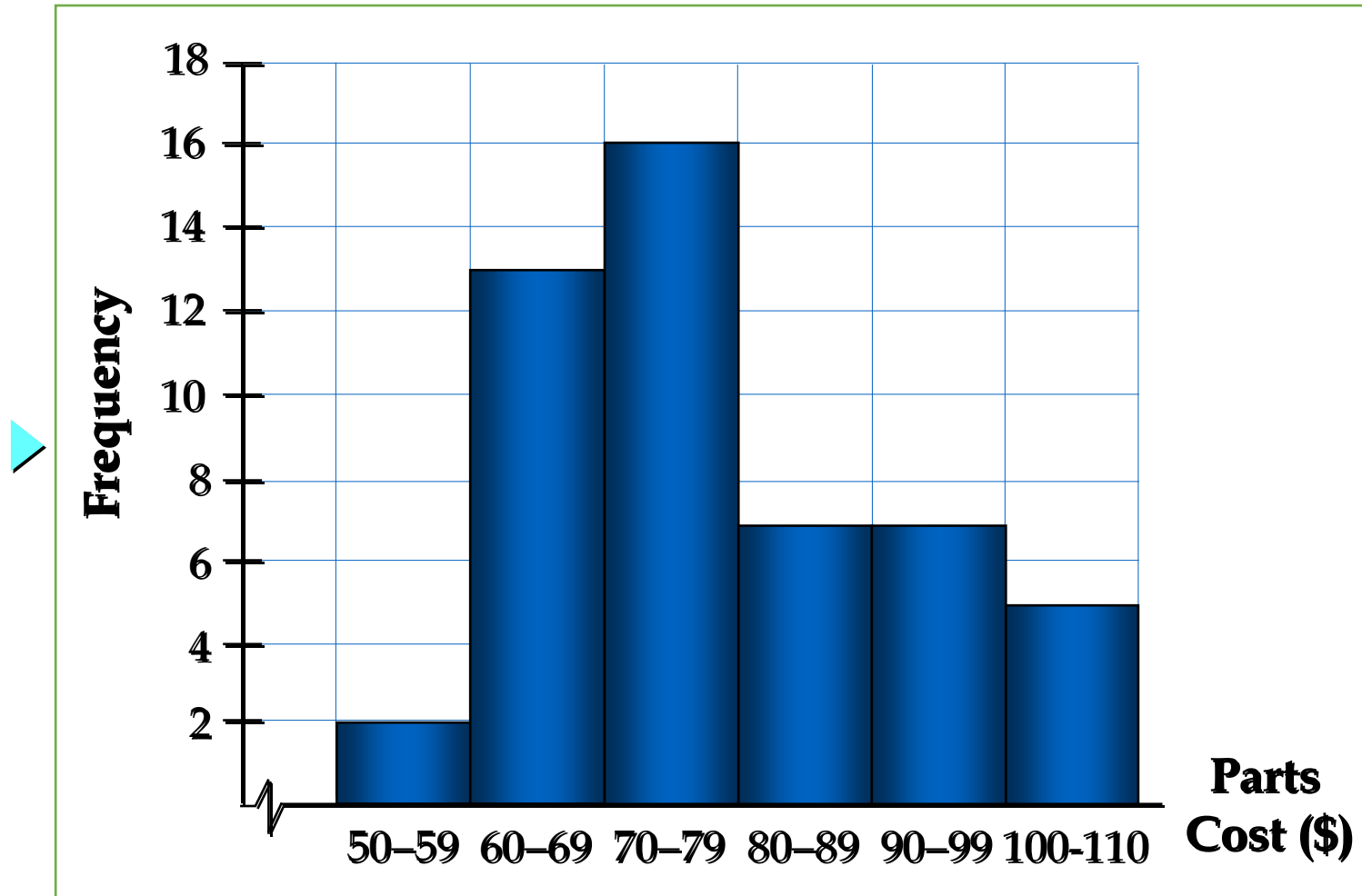
Another common graphical presentation of quantitative data is a histogram.

The variable of interest is placed on the horizontal axis.

A rectangle is drawn above each class interval with its height corresponding to the interval's frequency, relative frequency, or percent frequency.

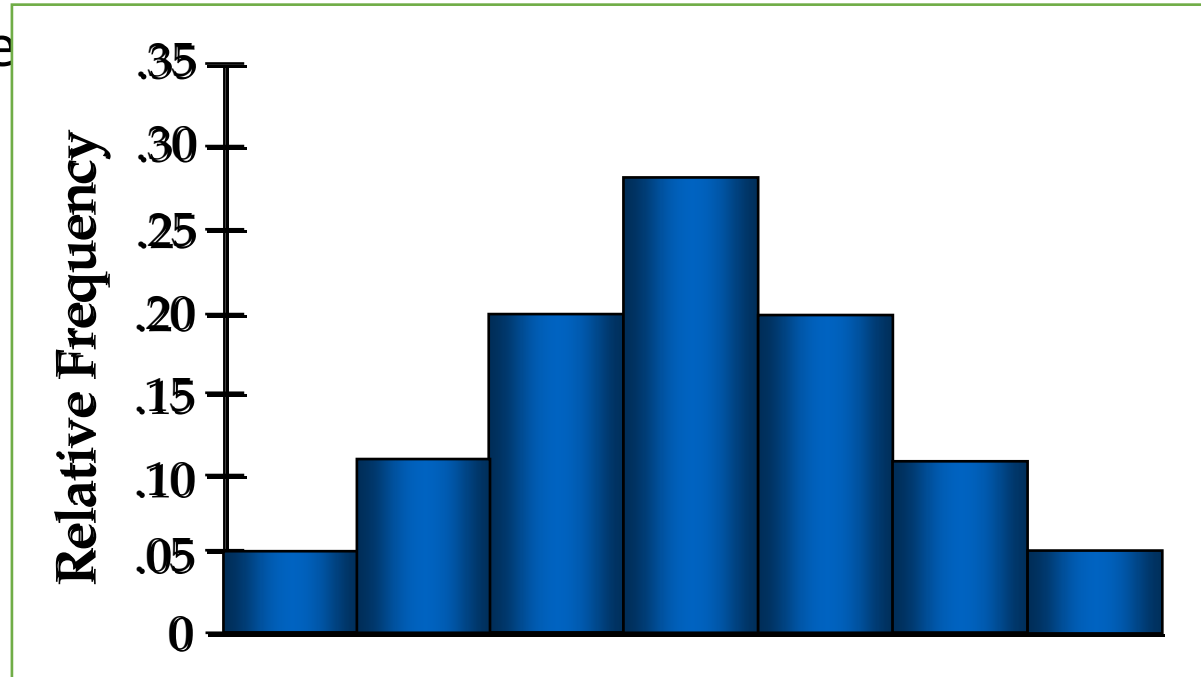
Unlike a bar graph, a histogram has no natural separation between rectangles of adjacent classes.

Histogram



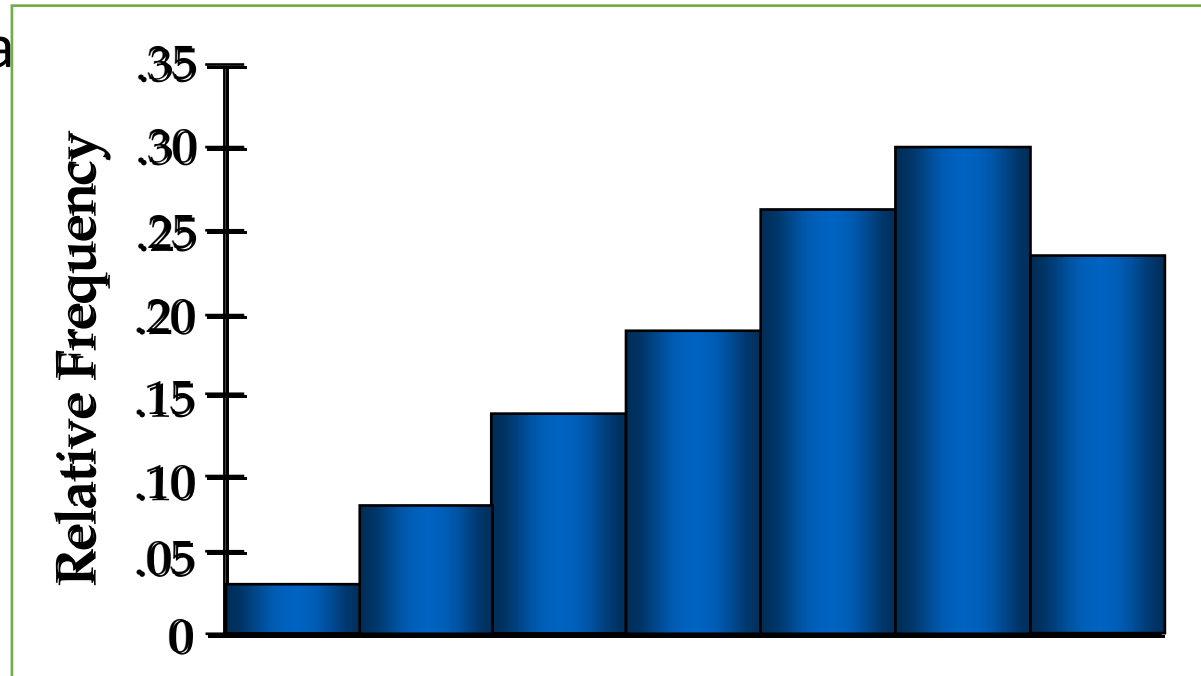
Histogram

- Symmetric
 - Left tail is the mirror image of the right tail
 - Examples: height



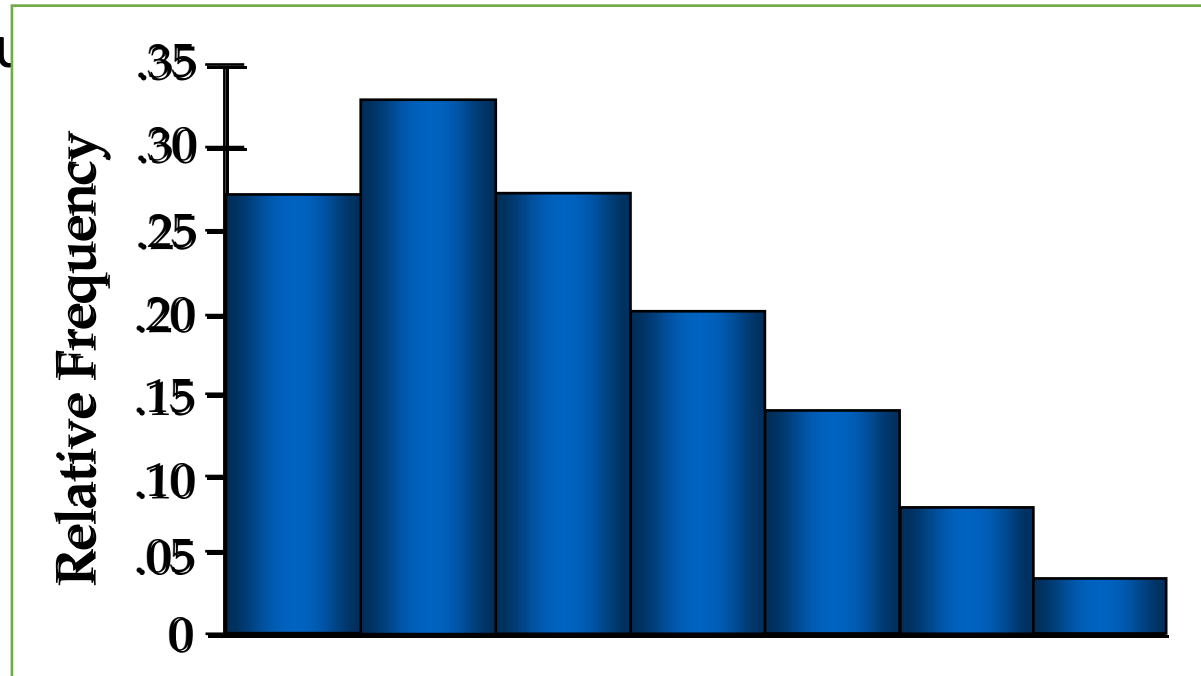
Histogram

- Skewed Left (negatively skewed)
 - A longer tail to the left
 - Example: exam



Histogram

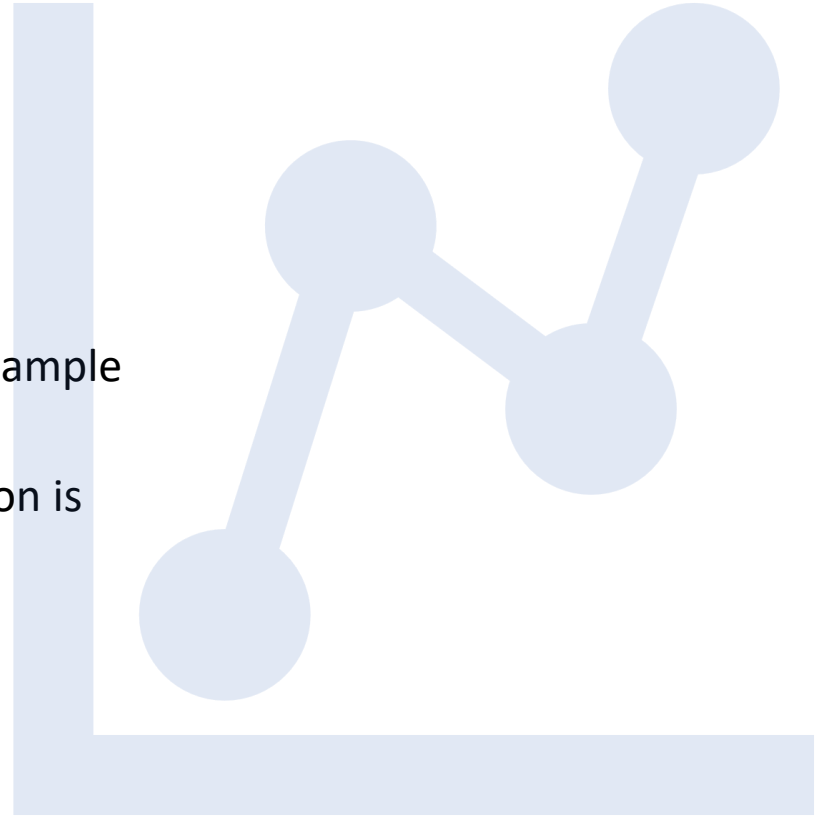
- Right Skewed (positively skewed)
 - A Longer tail to the right
 - Example: hou



DESCRIPTIVE STATISTICS for Quantitative variables: Measures of central tendency

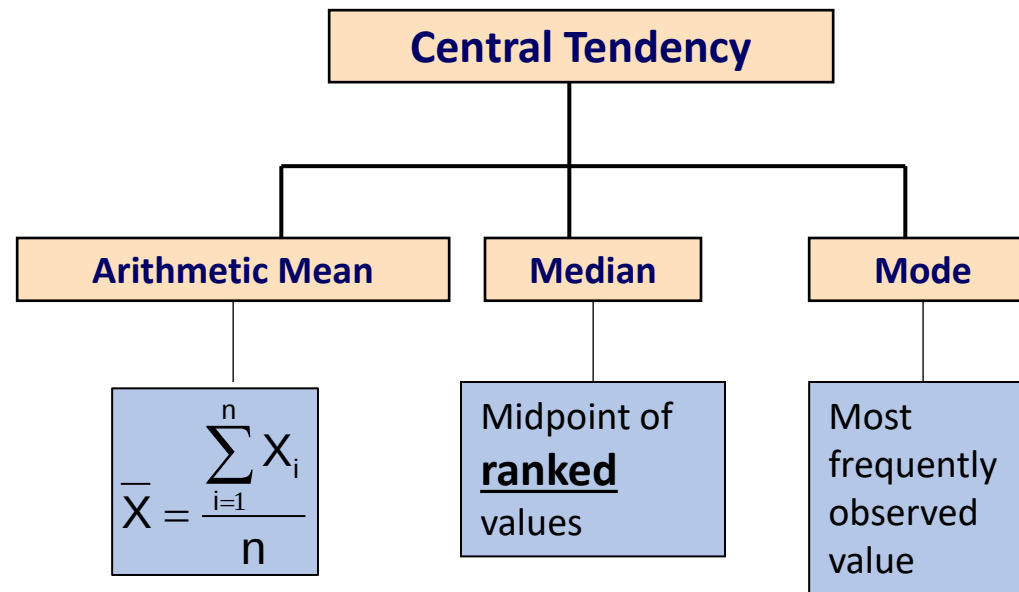


- Descriptive measures computed from the data of the sample are called ***statistics***.
- Descriptive measures computed from data of population is called ***parameter***.
- **Measures of central tendency:**
 - Mean
 - Median
 - mode



Measures of Central Tendency

Overview



Arithmetic Mean

- The arithmetic mean (mean) is the most common measure of central tendency
- For a sample of size n:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

The diagram illustrates the formula for the arithmetic mean. It features two orange boxes at the bottom. The box on the left, labeled "Sample size", has an orange arrow pointing to the denominator 'n' in the first fraction of the formula. The box on the right, labeled "Observed values", has an orange arrow pointing to the sum of values $X_1 + X_2 + \dots + X_n$ in the numerator of the second fraction.

Arithmetic Mean

- Example :

For the following values find the sample mean

$$S=\{4, 5, 7, 10, 9\}$$

First find

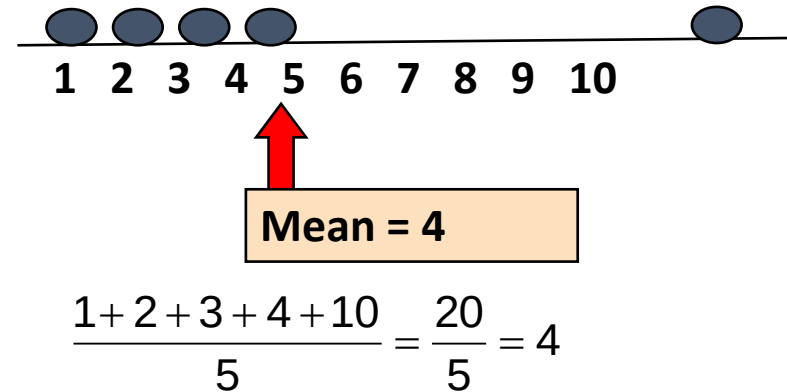
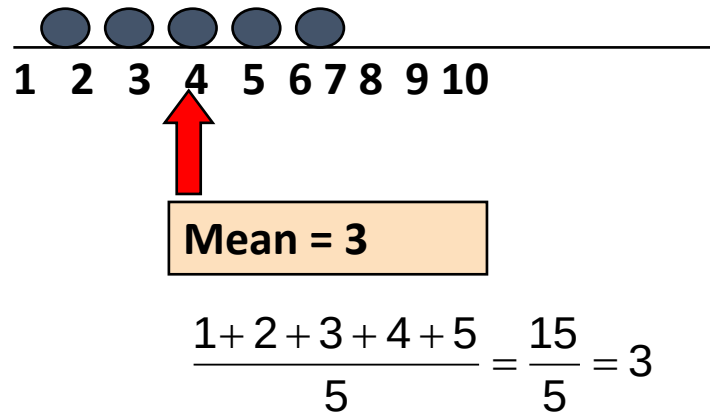
$$n=5$$

Then

$$\bar{x} = \frac{4+5+7+10+9}{5} = 7$$

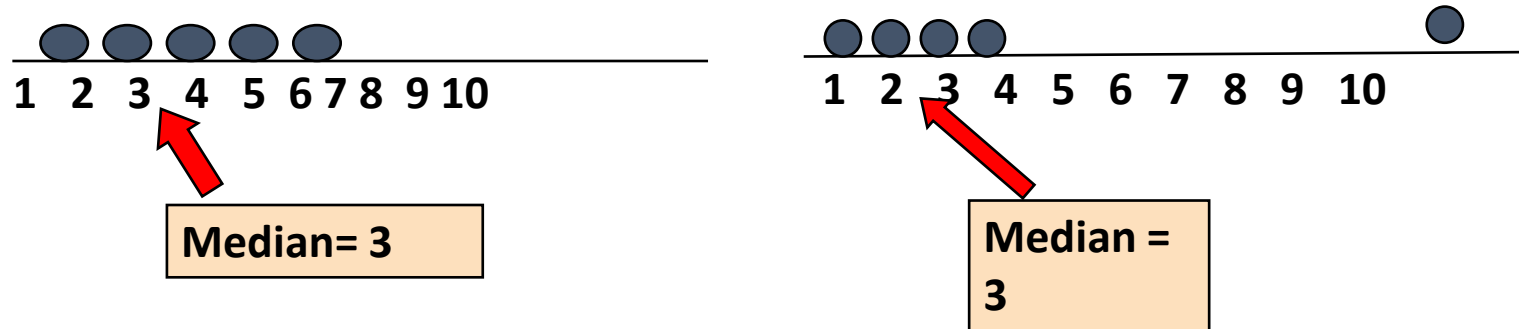
Arithmetic Mean

- The most common measure of central tendency
- Mean = sum of values divided by the number of values
- Affected by extreme values (outliers)



Median

- In an ordered array, the median is the “middle” number (50% above, 50% below)



- Not affected by extreme values

Median

- Data should be arranged in either ascending or descending array.
- Properties of median :
 - Uniqueness.
 - Simplicity.
 - Not affected by extreme value.

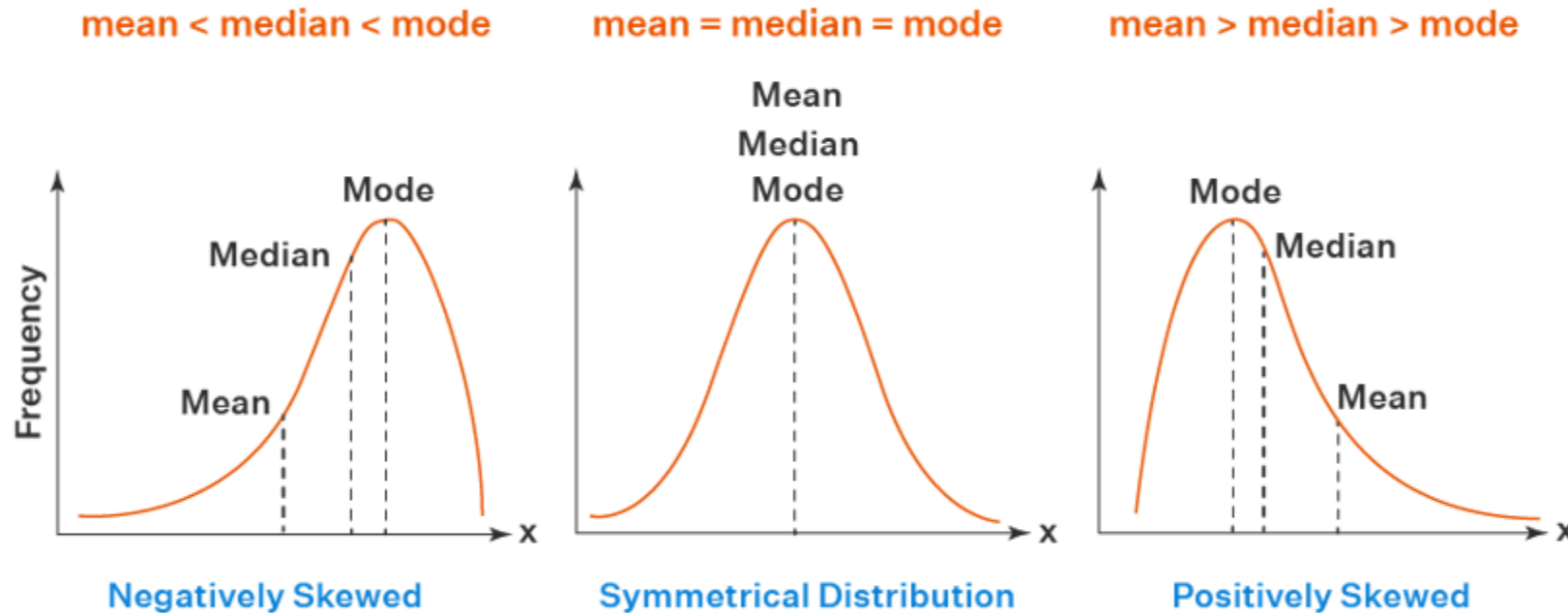
Mode

- **Mode is the value which occurs most frequently.**

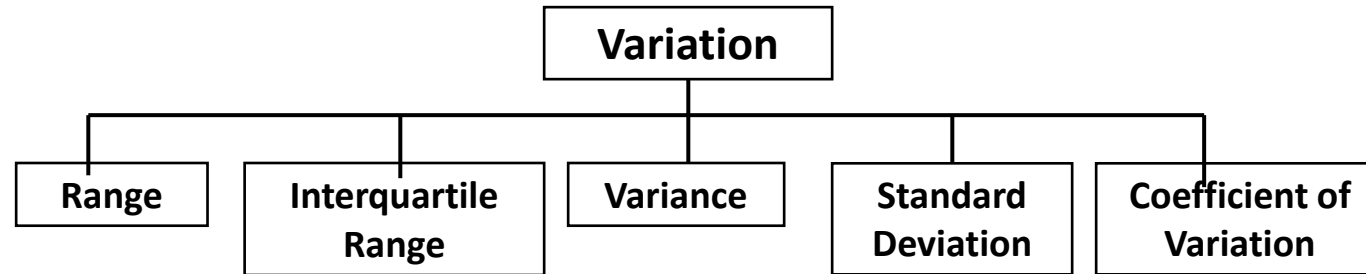
Either there is:

- **One mode** (the most common values).
- **Two or more modes** (if two or more values are "tied" for being the most common).
- **No mode** (all the values have the same frequencies).

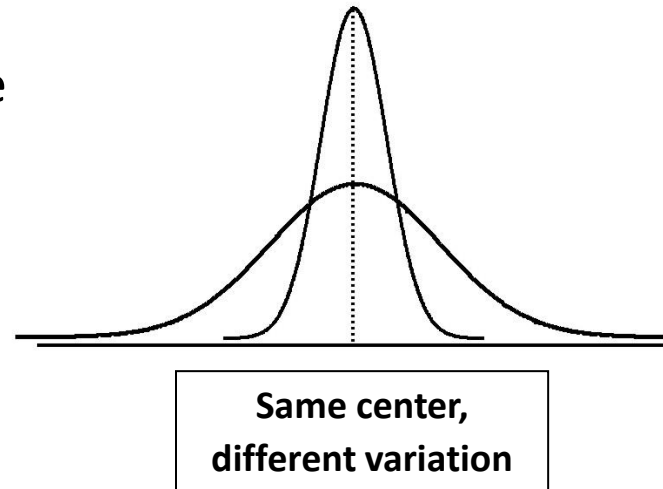
Relation Between Mean Median and Mode



Measures of Variation



Measures of variation give information on the **spread** or **variability** of the data values.



Range

- Is the different between the largest and smallest value in set of observation (one value).
- $R = X_l - X_s$
- Advantage: simplicity, but its poor measure of dispersion.

Example: 1,2,3,4,5

The range = $5 - 1 = 4$



Variance

- Average (approximately) of squared deviations of values from the mean

- Sample variance:

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

Where:

$\bar{X}$ = arithmetic mean

n = sample size

X_i = i^{th} value of the variable X

Standard Deviation

- Most commonly used measure of variation
- Shows variation about the mean
- Has the same units as the original data

- Sample standard deviation:

$$S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

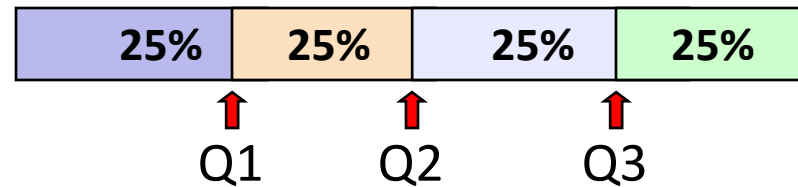
Measures of Position – Quartiles and Percentiles

Position

- Used to describe the position of a data value **in relation to the rest of the data.**
- Types:
 1. Quartiles
 2. Percentiles

Quartiles

- Quartiles split the ranked data into 4 segments with an equal number of values per segment



The first quartile, Q_1 , is the value for which 25% of the observations are smaller and 75% are larger

Q_2 is the same as the median (50% are smaller, 50% are larger)

Only 25% of the observations are greater than the third quartile

Example.....

- Find Q1, median, and Q3.
- 1, 3, 10, 12, 13, 15, 17, 28, 30, 34, 41 is the data ordered?

Find the median (Q2)

1, 3, 10, 12, 13, 15, 17, 28, 30, 34, 41

Then find Q1

1, 3, 10, 12, 13

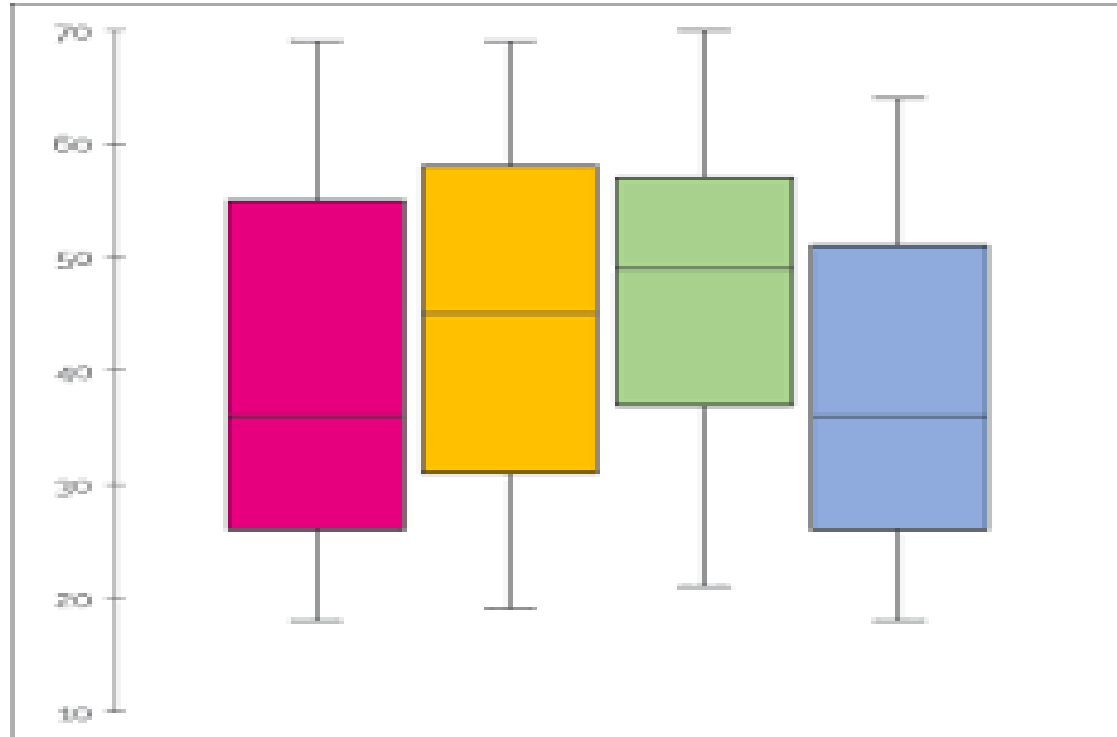
Then find Q3

17, 28, 30, 34, 41

Boxplot

- A graph of a set of data obtained by drawing a horizontal line from the minimum to maximum values with quartiles labeled in between.
- It is a graphical plot of 5 specific values called the **5-number summary**.

Box Plots (Box and Whiskers)



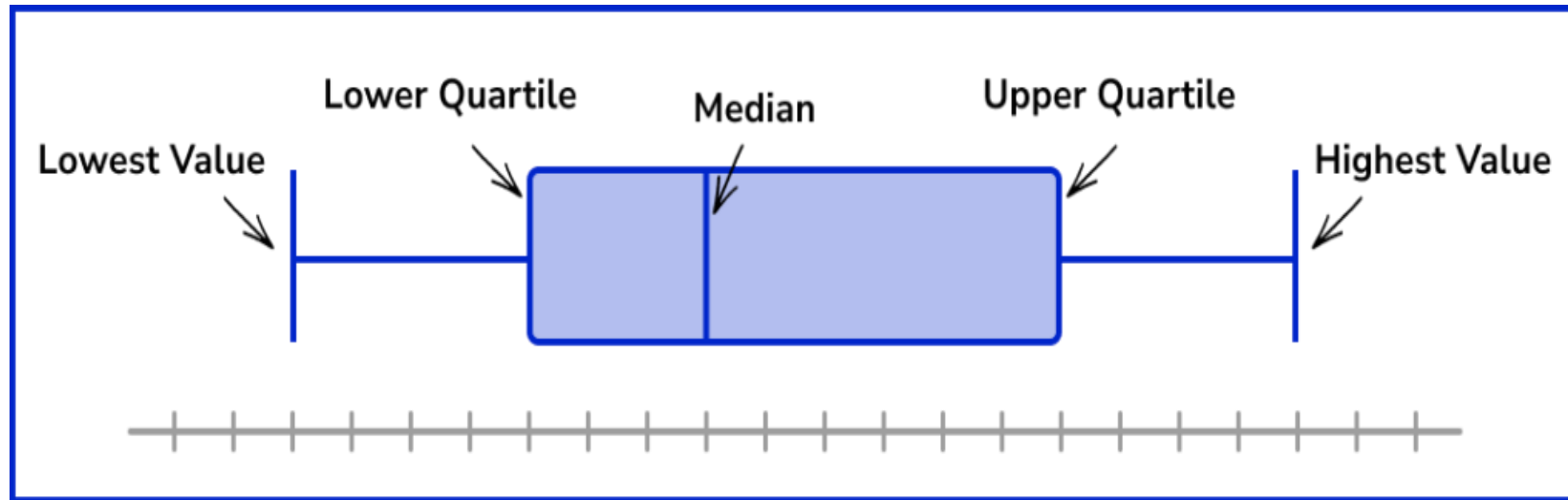
How do we interpret a box plot?

- It is a valuable source of easy-to-interpret information about a sample of study.
- It can provide information about a sample's range, median, normality of the distribution, and skew of the distribution.

5-number summary.....

- 1. minimum
- 2. Q1
- 3. median
- 4. Q3
- 5. maximum

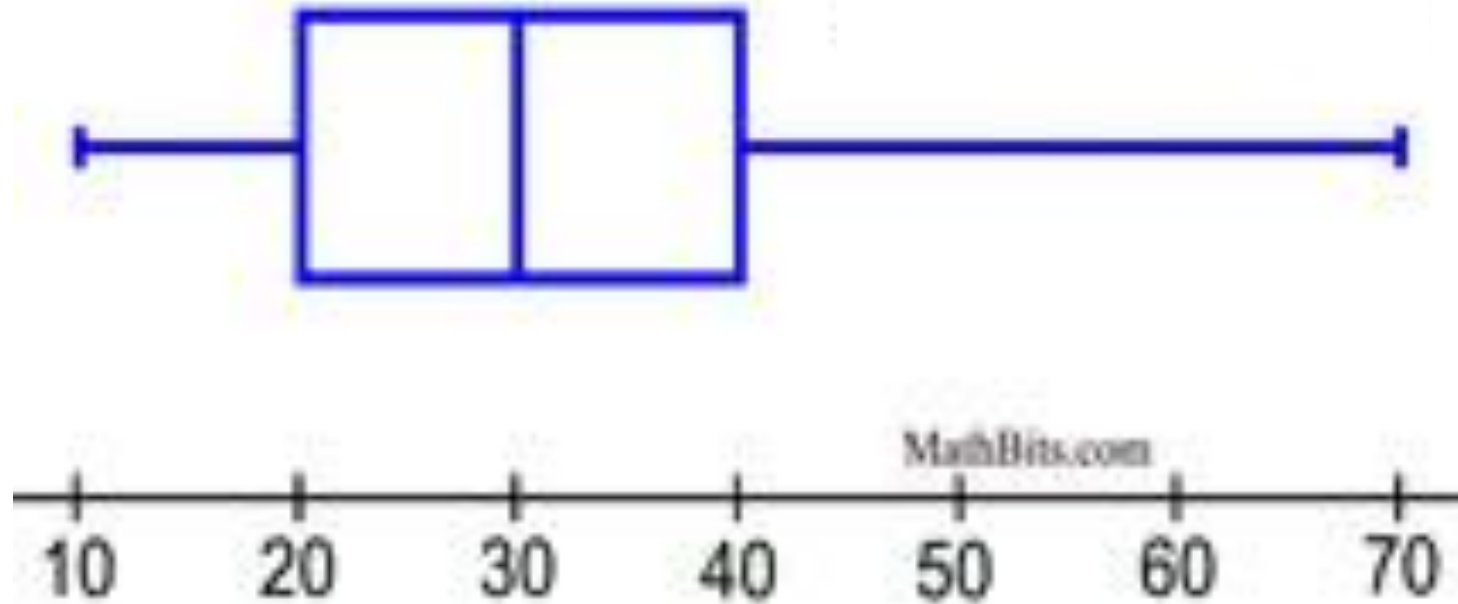
Box plot



Skewness

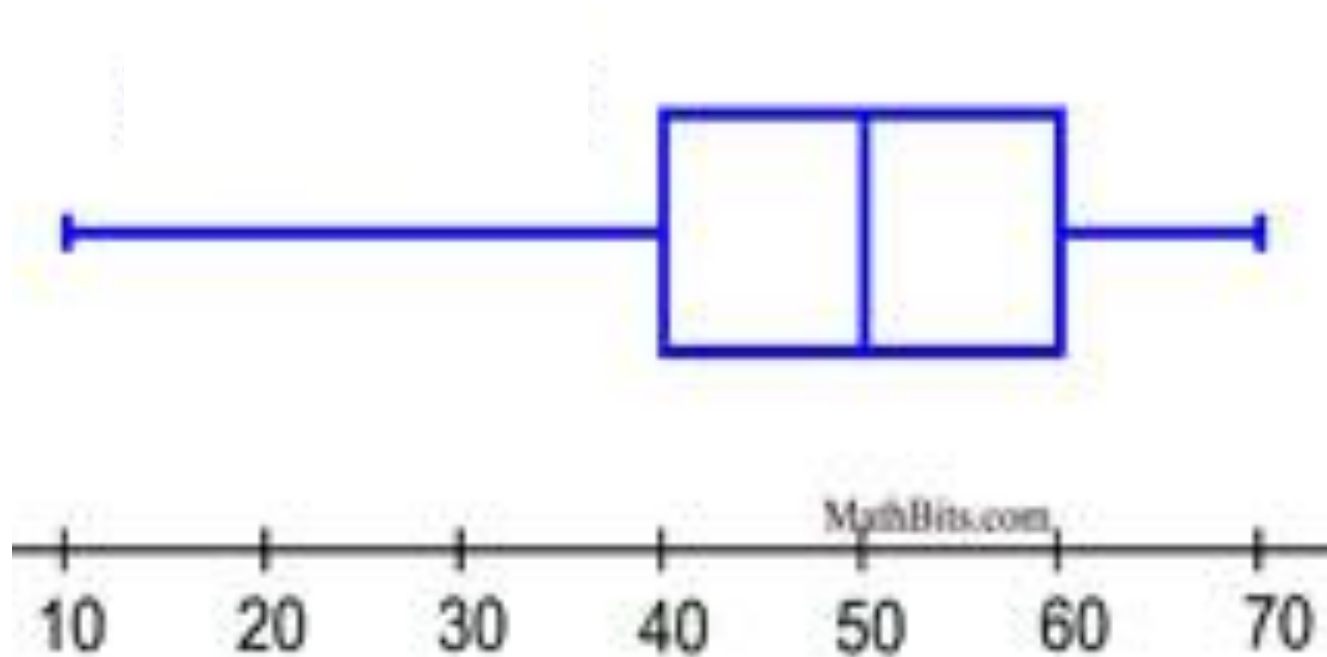
- If the whisker to the right of the box is longer than the one to the left in relation to the median, there are more extreme values towards the positive end and so the distribution is **positively skewed**.
- Similarly, if the whisker to the left is longer, the distribution is **negatively skewed**.

Skewed Left (Negative) or Right (Positive)?



Right Skewed

Skewed Left (Negative) or Right (Positive)?



Left Skewed